

LACONIA, NH, USA

WMO: 726155

Lat: 43.567N

Lon: 71.433W

Elev: 545

StdP: 14.41

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 54736

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.1	(c) 1.4	(d) -17.6	(e) 2.1	(f) 0.7	(g) -11.2	(h) 3.1	(i) 5.0	(j) 23.0	(k) 27.2	(l) 20.5	(m) 24.8	(n) 6.4	(o) 260	(p) 0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.1	(c) 88.2	(d) 71.5	(e) 84.3	(f) 69.5	(g) 81.8	(h) 67.7	(i) 73.8	(j) 83.6	(k) 72.1	(l) 81.2	(m) 70.4	(n) 78.8	(o) 9.5	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.5	114.5	78.6	69.6	111.1	77.4	67.1	101.6	75.4	37.7	83.7	36.1	81.1	34.6	79.0	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 19.6	(b) 17.4	(c) 15.6	(e) -9.0	(f) 92.9	(g) 5.6	(h) 2.7	(i) -13.1	(j) 94.8	(k) -16.4	(l) 96.4	(m) -19.6	(n) 97.9	(o) -23.7	(p) 99.8		
(4) 19.6	(5) 17.4	(6) 15.6	(7) -9.0	(8) 92.9	(9) 5.6	(10) 2.7	(11) -13.1	(12) 94.8	(13) -16.4	(14) 96.4	(15) -19.6	(16) 97.9	(17) -23.7	(18) 99.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.6	21.1	23.5	32.3	44.2	55.4	65.1	70.5	68.6	60.9	49.2	38.2	28.2
	DBStd	18.84	10.75	9.87	9.32	8.08	7.49	6.81	5.42	5.29	7.00	7.56	8.43	8.95
	HDD50	3586	896	743	553	213	28	1	0	0	4	108	364	676
	HDD65	7189	1361	1163	1012	626	313	82	12	24	162	490	803	1141
	CDD50	2327	0	0	5	38	195	453	636	575	330	84	11	0
	CDD65	457	0	0	0	1	15	84	183	134	38	2	0	0
	CDH74	3852	0	0	8	41	225	760	1519	1019	264	16	0	0
	CDH80	1066	0	0	1	10	61	224	460	256	54	0	0	0
	WSAvg	6.2	6.7	6.8	6.8	7.0	6.4	5.8	5.5	5.0	5.2	6.2	6.4	6.7
	PrecAvg	46.0	3.1	3.1	3.9	3.7	3.6	4.1	4.1	3.9	3.7	4.5	4.3	4.1
(15) Precipitation	PrecMax	60.7	6.4	8.7	7.0	7.8	9.6	11.7	7.9	8.5	9.5	14.5	6.2	8.0
	PrecMin	30.6	0.6	1.5	1.5	0.4	0.4	1.2	0.5	0.9	1.1	0.6	1.9	1.7
	PrecStd	7.9	1.3	1.5	1.4	1.9	2.0	2.5	1.8	2.2	2.0	3.3	1.0	1.5
	PrecStd	7.9	1.3	1.5	1.4	1.9	2.0	2.5	1.8	2.2	2.0	3.3	1.0	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	54.5	54.2	69.5	80.6	86.4	90.9	91.2	90.4	85.8	76.6	65.8	56.5	52.3
	0.4% MCWB	51.9	45.5	56.2	60.1	65.6	72.8	73.9	72.2	70.7	63.4	60.1	50.2	50.2
	2% DB	44.7	45.9	56.6	71.9	80.9	86.1	88.3	85.7	81.0	71.5	61.2	48.5	48.5
	2% MCWB	39.8	38.8	46.0	55.6	62.1	69.9	72.2	70.8	67.5	61.1	54.2	44.0	44.0
	5% DB	38.8	41.1	50.0	64.3	74.9	81.7	84.4	82.2	76.7	66.0	55.3	44.7	44.7
	5% MCWB	34.6	36.0	41.1	51.4	59.6	66.9	70.0	68.7	65.3	58.3	49.1	40.1	40.1
	10% DB	35.9	36.8	45.4	59.1	70.4	78.7	81.8	80.6	72.9	62.3	51.8	39.5	39.5
	10% MCWB	32.5	32.5	38.3	47.1	57.2	64.5	68.5	67.4	63.4	54.6	45.9	35.4	35.4
	0.4% WB	52.5	47.4	56.2	63.5	68.9	75.0	76.5	75.2	73.1	67.0	61.5	52.3	52.3
	0.4% MCDB	53.9	50.7	67.0	73.3	82.2	86.1	87.6	84.6	80.6	71.8	65.2	55.1	55.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	40.3	40.4	47.8	57.9	65.5	72.4	74.1	72.8	70.0	63.0	55.8	45.4	45.4
	2% MCDB	43.7	45.0	55.2	69.1	75.7	83.1	84.2	81.3	77.1	68.1	59.9	48.1	48.1
	5% WB	35.3	36.5	42.3	52.9	62.2	69.7	72.4	71.1	67.8	59.1	50.5	40.4	40.4
	5% MCDB	38.1	40.5	48.8	62.5	72.1	78.2	81.8	79.2	74.0	65.1	55.2	44.0	44.0
	10% WB	32.4	33.1	38.6	48.5	59.1	67.2	70.6	69.4	65.7	55.5	46.3	36.9	36.9
	10% MCDB	35.1	36.7	44.4	58.0	67.8	74.9	79.4	76.7	71.6	60.6	49.7	39.7	39.7
(35) Mean Daily Temperature Range	MDBR	15.8	17.4	17.5	19.7	20.2	19.1	19.1	19.4	19.1	17.5	14.7	13.7	13.7
	5% DB MCDBR	18.6	20.8	25.3	30.5	28.4	24.1	22.1	22.2	22.4	23.8	21.4	18.2	18.2
	5% DB MCWBR	16.4	15.9	16.8	17.2	14.1	10.9	9.4	10.2	11.5	14.5	16.2	15.1	15.1
	5% WB MCDBR	18.0	20.0	23.1	27.9	23.7	21.1	19.3	18.9	18.6	20.6	19.1	17.6	17.6
	5% WB MCWBR	16.5	16.1	16.9	17.9	13.3	10.9	9.4	9.8	11.2	14.4	16.3	15.5	15.5
	5% WB MCWBR	16.5	16.1	16.9	17.9	13.3	10.9	9.4	9.8	11.2	14.4	16.3	15.5	15.5
(40) Clear-Sky Solar Irradiance	taub	0.275	0.289	0.316	0.348	0.379	0.427	0.445	0.414	0.366	0.331	0.308	0.283	0.283
	taud	2.421	2.366	2.417	2.423	2.374	2.254	2.213	2.289	2.406	2.512	2.531	2.488	2.488
	Ebn at Noon	275	289	294	291	283	269	262	267	273	271	261	261	261
	Edh at Noon	23	30	32	35	38	43	44	39	32	26	22	20	20
(44) All-Sky Solar Radiation	RadAvg	525	800	1150	1434	1593	1720	1795	1632	1327	833	558	413	413
	RadStd	39	64	72	161	190	158	124	89	81	83	56	43	43

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (63 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+144	N/A

Nomenclature: See separate page